

Project Olympus · Unit 0: Character Creation

Student practice sheets. Paper first. · The Language of Everything · David Balyeat, with Claude

Week 1 · Day 1: Welcome to Olympus

The rules of the game, and a trick that is secretly the whole course

Warm-up

No calculator. Is 19×21 closer to **400** or **4,000**?

Practice

1. Without a calculator: 48×52 is closest to which?

- A. 250
 - B. 2,500
 - C. 25,000
 - D. 250,000
-

2. The number of seconds in a day is closest to which order of magnitude?

- A. 1,000
 - B. 10,000
 - C. 100,000
 - D. 1,000,000
-

3. A soccer pitch is about 105 m long and 68 m wide. Hudson multiplies and gets an area of **71,400 m²**. Is that reasonable?

- A. Reasonable
 - B. Too big by about 10×
 - C. Too small by about 10×
 - D. Too big by about 100×
-

4. A new version of the trick: think of a number, **triple it, add 12, divide by 3**, then **subtract your original number**. What does everyone get?

5. In the original trick (double, add 10, halve, subtract original), the magician changes "add 10" to "add 20." Now what's the answer everyone gets?

6. Hero rank is **1,500 XP**. Suppose a session earns about **90 XP**. About how many sessions to Hero?

7. Spider-Man web-swings across Queens at roughly **25 m/s** (about 55 mph; the movies are vague, so we're estimating). About how many seconds to cover **1 kilometer**?

Week 1 · Day 2: Fermi's Napkin

How physicists estimate anything in 60 seconds

Warm-up

Powers of ten: $10^3 \times 10^2 = ?$ (Type the plain number.)

Practice

1. How many **seconds are in a year**? Estimate it (no calculator until you've guessed), then calculate.

2. How many times has your **heart beaten** in your life so far? (You're 13. Resting heart rate is about 70 beats per minute; you've been asleep, running, and sitting, so 70 is a fair average.)

3. How many **kicks of a soccer ball** happen worldwide in one day? (FIFA estimates about 250 million people play soccer regularly.)

4. How many **jiujitsu sparring rounds** are happening worldwide during one of your one-hour classes? (Rough guess: about 3 million people train BJJ worldwide.)

5. How many **Lego bricks** would a **life-size Millennium Falcon** need, if you built it solid? The Falcon is about 35 m long, 25 m wide, and 8 m tall (but it's a saucer, not a box: use about 1/3 of the box volume). A standard 2×4 brick is 3.2 cm × 1.6 cm × 1 cm.

6. Calculate: $(3 \times 10^4) \times (2 \times 10^3)$. Type the plain number.

7. Your Fermi estimate for something was 4×10^8 . The true value turned out to be 5.2×10^8 . By what factor were you off? (Divide the bigger by the smaller.)

Week 1 · Day 3: The Three Tools

Calculator, Sheets, and AI: how each one lies to you, and how to catch it

Warm-up

No calculator: $2 + 3 \times 4 = ?$

Practice

1. Estimate first, then use the calculator: $20 - 8 \div 2 \times 3$

2. $2 + 3^2 \times (4 - 1)$

3. In a Sheet, cell A1 holds **12** and B1 holds **5**. What does `=A1*B1+2` show?

4. $A1 = 100$, $A2 = 200$, $A3 = 300$. What does `=AVERAGE(A1:A3)` show?

5. What does `=2^10` show?

6. An AI tells you: "Lionel Messi's top speed at the 2026 World Cup was 40 km/h." Sofascore's recorded figure was 30.9 km/h. What's the right move?

- A. Trust the AI; it's usually right
 - B. Check the source, record the AI's mistake, and claim the Bug Hunter badge
 - C. Average the two numbers
 - D. Ask the AI again until it agrees with you
-

7. Use the $\times 10^n$ key: $(4 \times 10^6) \div (8 \times 10^2)$. Type the plain number.

Week 1 · Day 4 · Lab Day: Lab Day: The First Labor

Labor 0, Quest 1, and the trick goes public

Warm-up

Head math: **15% of 80?**

Practice

1. A quick Fermi check before the Labor: the real Millennium Falcon is about **34.75 m** long. The big Lego set (75192) is about **84 cm** long. Roughly how many times longer is the real ship? (One significant figure is fine.)

2. If every *length* gets multiplied by 41, what does the *volume* get multiplied by? (Think of a cube: length $\times$ width $\times$ height, each one 41 $\times$ bigger.)

3. Bug Hunter practice. An AI says: "41 cubed is 68,291." Is it right?

- A. Yes
 - B. No, it's 68,921 (digits swapped)
 - C. No, it's 6,892
-

Labor 0: The Fermi Falcon

On Day 2 you estimated the bricks in a life-size Millennium Falcon from its volume. Now check it a completely different way: scale up the real Lego set. If two methods agree, you can trust them both. No hints. Paper, calculator, Sheets allowed.

Part 1. Part 1. The Lego UCS Millennium Falcon (set 75192) is about **84 cm** long. The "real" Falcon is about **34.75 m**. Find the scale factor (how many times longer the real one is). Two decimal places.

Part 2. Part 2. If each length is $41.37\times$ longer, volume is 41.37^3 times bigger. Compute that. (Nearest thousand is fine.)

Part 3. Part 3. The set has **7,541 pieces**. If you scaled every piece up, you'd need $7,541 \times 41.37^3$ pieces of *giant* Lego. But you want *normal* bricks, so that's your brick count. Compute it in scientific notation (e.g. $3.2e8$).

Part 4. Part 4. Lego averages roughly **10 cents per piece**. What would the bricks cost, in dollars? (Scientific notation or plain number.)
